



Introduction to BESS Availability Types

How to measure BESS performance

Module 1 - BESS Availability Explained

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From OEE to OBE

OEE – Overall Equipment Effectiveness

(Manufacturing)

- **Availability** → Is the machine on?
- **Performance** → Is it running at full speed?
- **Quality** → Is it making good parts?

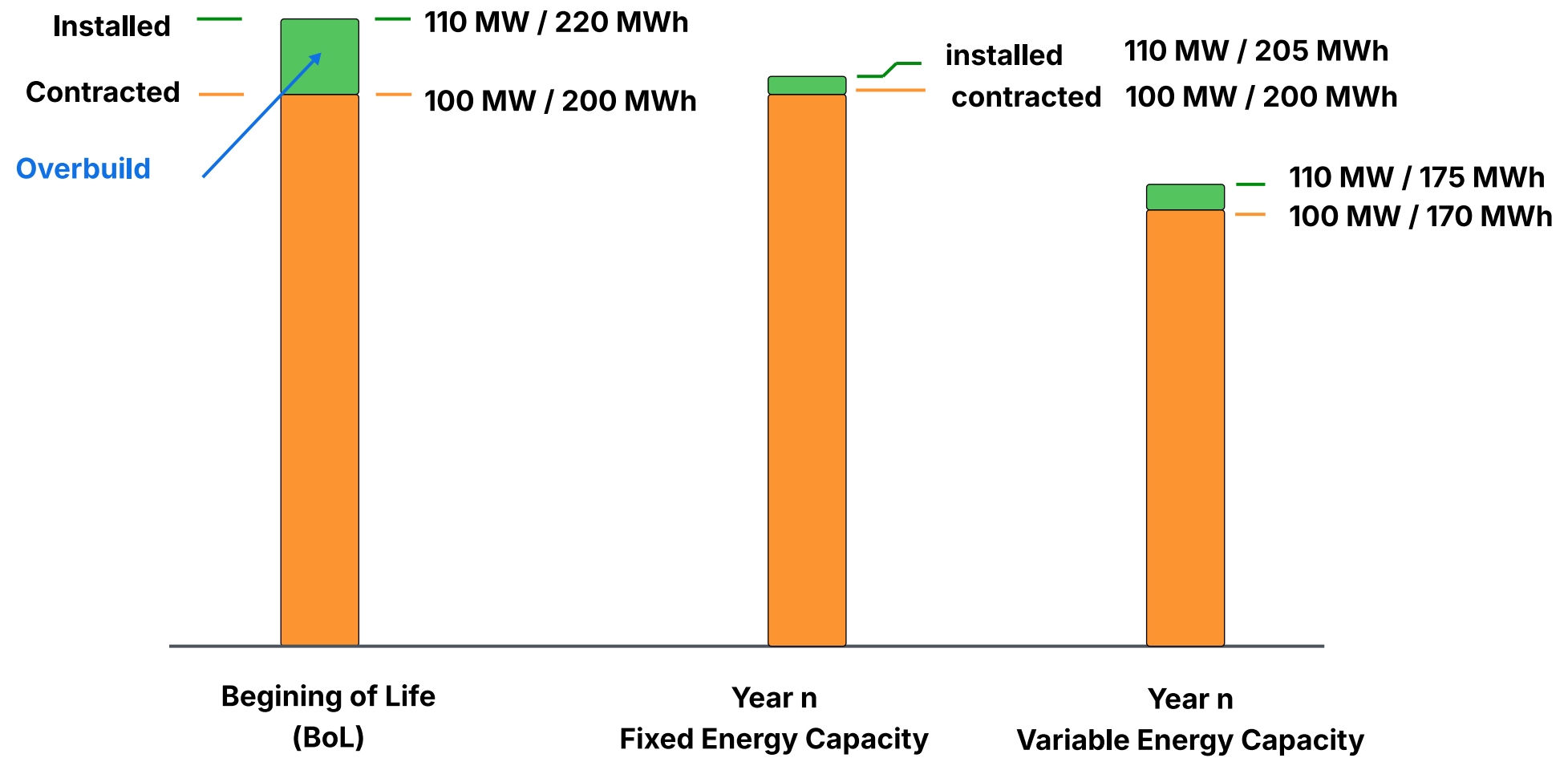
OBE – Overall Battery Effectiveness

(BESS)

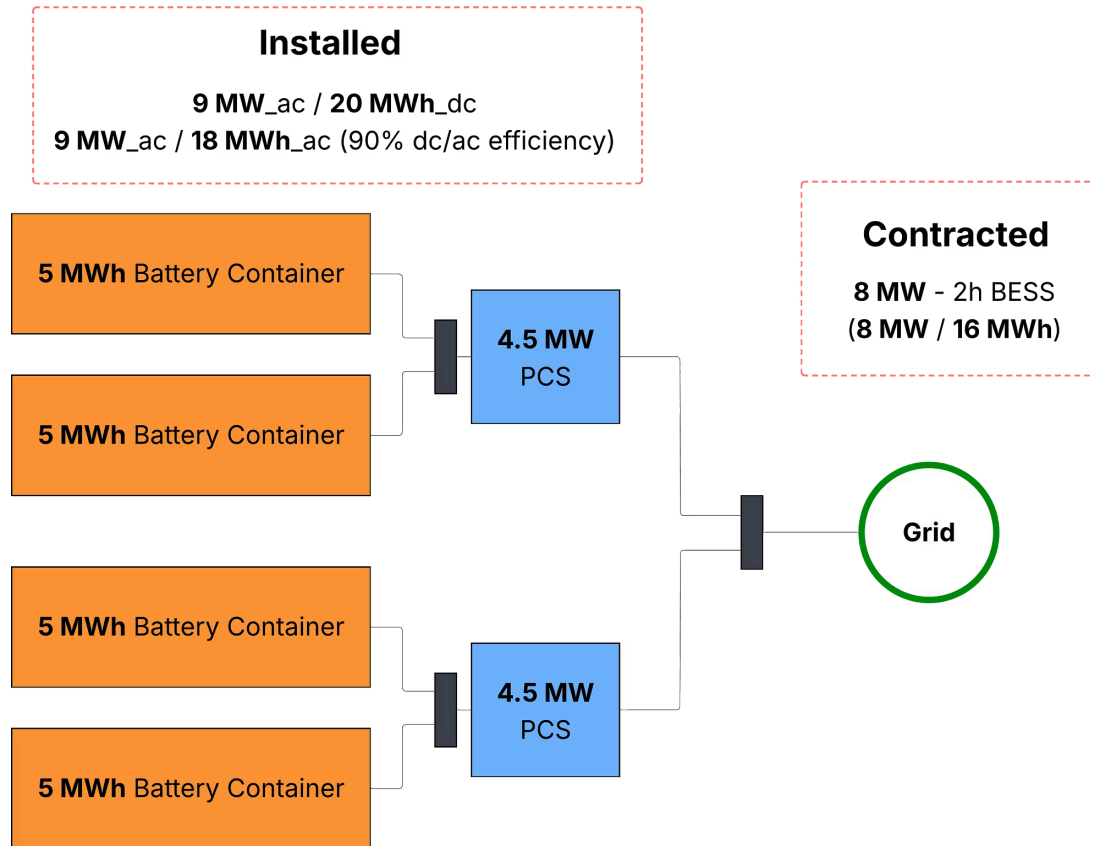
- **Equipment Availability** → Is the BESS on?
- **Operational Availability** → Is it responding to dispatch?
- **Power Availability** → Is it delivering full power?
- **Energy Availability** → Is it delivering good energy?

OBE = the combination of all four BESS availability types

BESS Capacity Types



Installed vs Contracted Install Example (8 MW 2h BESS)



① Equipment Availability

<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary_</u> <u>Unit</u>	<u>Secondary Unit</u>
MEDIUM (ON/OFF)	HIGH 100% of Calendar Time	HIGH (Equipment Status signals)	HIGH Imbalance, SoC error, Derates	%	MW_installed MW_contracted

② Operational Availability

<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
HIGH (MW)	LOW Only During Dispatch	HIGH (Meter Measurement/ Setpoint)	HIGH All time with dispatch command < nameplate power	MW	%_contracted

③ Power Availability

<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
HIGH (MW)	HIGH 100% of Calendar Time	MEDIUM Calculated by EMS. Based on various equipment reported metrics	MEDIUM Many corner case and special condition that might be hard to model.	MW	%_installed %_contracted

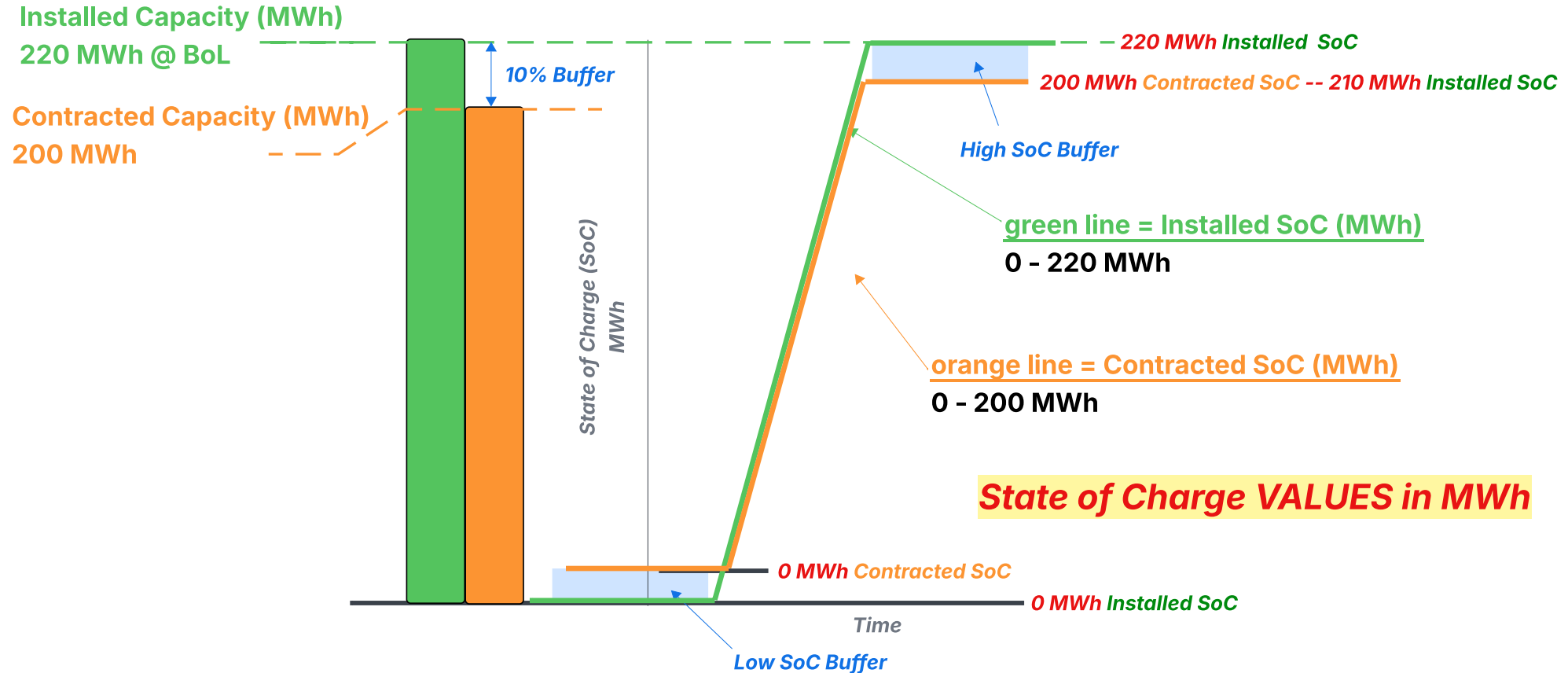
④ Energy Availability

<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
HIGH (MWh)	MEDIUM 100% of Calendar Time but might only be updated with capacity tests annually	LOW to MEDIUM Calculated by EMS, include SoC Error and Imbalance	MEDIUM Might include power derated energy due to imbalance	MWh	%_installed %_contracted

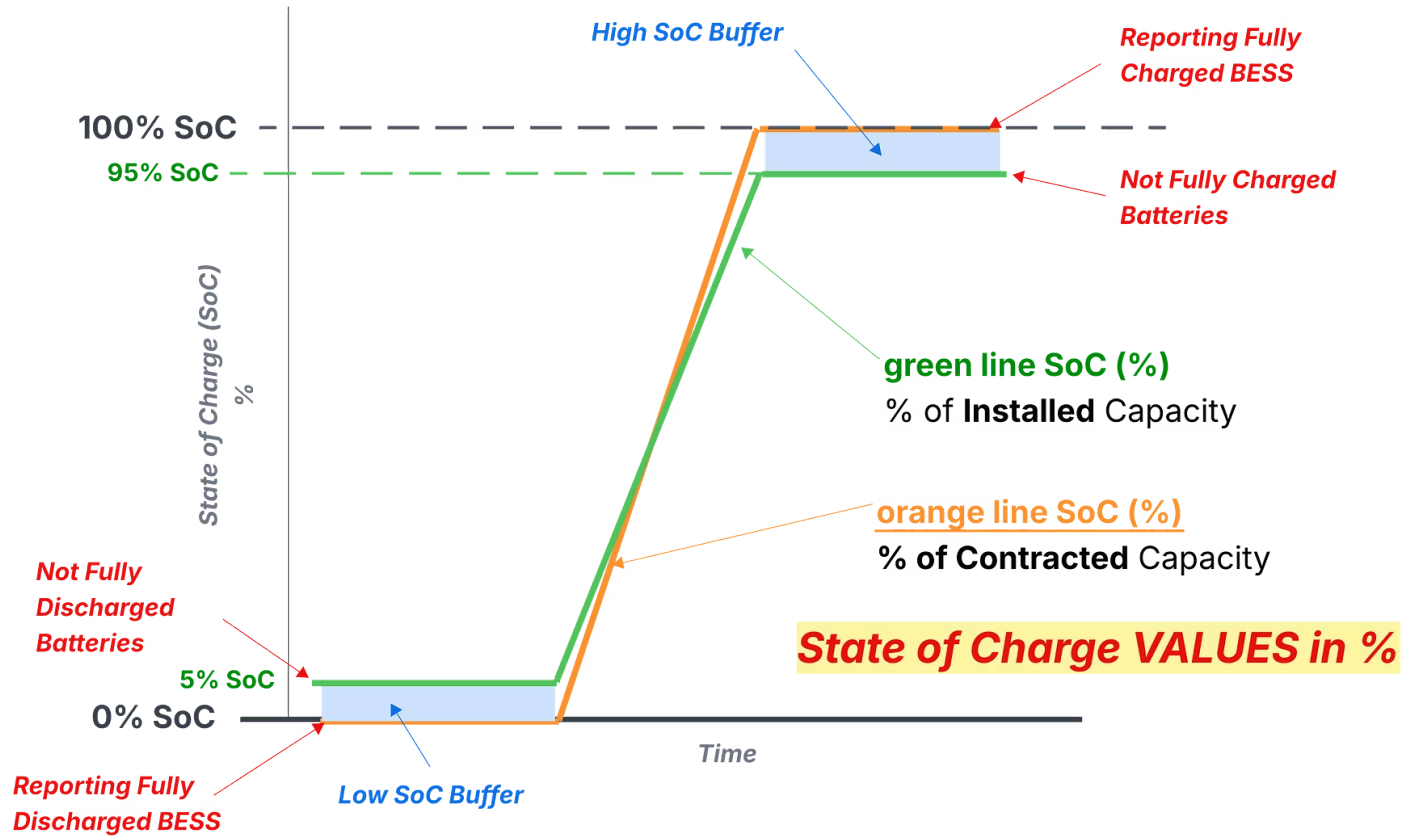
BESS Availability Types Recap

<u>Availability Type</u>	<u>Granularity</u>	<u>Coverage</u>	<u>Accuracy</u>	<u>Blind Spots</u>	<u>Primary Unit</u>	<u>Secondary Unit</u>
① Equipment Availability	MEDIUM (ON/OFF)	HIGH 100% of Calendar Time	HIGH (Equipment Status signals)	HIGH Imbalance, SoC error, Derates	%	MW_installed MW_contracted
② Operational Availability	HIGH (MW)	LOW Only During Dispatch	HIGH (Meter Measurement/ Setpoint)	HIGH All time with dispatch command < nameplate power	MW	%_contracted
③ Power Availability	HIGH (MW)	HIGH 100% of Calendar Time	MEDIUM Calculated by EMS. Based on various equipment reported metrics	MEDIUM Many corner case and special condition that might be hard to model.	MW	%_installed %_contracted
④ Energy Availability	HIGH (MWh)	MEDIUM 100% of Calendar Time but might only be updated with capacity tests annually	LOW to MEDIUM Calculated by EMS, include SoC Error and Imbalance	MEDIUM Might include power derated energy due to imbalance	MWh	%_installed %_contracted

Installed vs Contracted MWh Capacities



Percent of Installed vs Contracted Capacity



OBE – Overall Battery Effectiveness

Like OEE in manufacturing but built for BESS

$$OBE = EA \times OA \times PA \times QA$$

EA

Equipment Availability

OA

Operational Availability

PA

Power Availability

QA

Energy Availability

At any given interval, OBE is constrained by the weakest of the four availability types, just as OEE is the product of its components.

What's Coming Next

We've covered the fundamentals of BESS availability and capacity types. In our next sessions, we'll dive deeper into each availability type, starting with a comprehensive look at **Equipment Availability**. Stay tuned for more Performance Engineering insights!

